

GovTech GASP: Job Scam Detection

Purpose

Use our analysis of a labeled job-post dataset (job text fields + basic posting flags) to produce actionable insights for a pre-publication job scam detection pilot that supports GovTech's public safety mission.

1) Data quality & limitations

- Missing fields are common, and missingness is meaningful (not random).
- Links appear as placeholders (e.g., #URL). We normalized these to a consistent token so we don't accidentally lose link-related signals.
- The "fraudulent" label is heterogeneous: it includes classic scam templates but also other fraud-like postings (e.g., recruitment spam/boilerplate). This is why we introduced scam-like subtyping before pattern discovery.

2) Key findings

A) Fraud ads are structurally different from legit ads

CDF plots show:

- Fraud ads tend to be shorter than legitimate ads (clear left-shift in the length distribution).
- Fraud ads are more number-heavy (digit density is shifted toward higher values), consistent with pay/earnings framing and conversion bait.

These are robust signals that do not depend on exact wording.

B) URL presence is not a strong standalone signal

has_url proportion chart shows similar URL rates in fraud and legit.

Interpretation:

Legitimate postings frequently include application links too. So "has a link" alone is not meaningful here. Links become useful only when combined with credibility gaps or bait signals

C) The strongest risk indicators are combinations (feature bundles), not single signals

My bundle table shows that fraud risk increases substantially when credibility gaps and bait framing co-occur, especially combinations involving:

- Missing company profile paired with:
 - high digit density
 - short text
 - pay-frequency framing (daily/weekly/per hour)
 - missing requirements / benefits
 - telecommuting

Interpretation:

Scam-like ads often look “thin” (low employer credibility) *and* contain strong conversion bait (pay framing / numbers / quick, light work). Together these are much more suspicious than any single feature.

D) Single-feature signals: credibility signals are strongest

From my odds ratio output:

- **Missing company_profile** is the single strongest “credibility gap” signal (much more common in fraud).
- **High digit density** and **short text** are strong supporting signals.
- **Telecommuting** leans fraud, but is a weaker signal on its own.
- **Has logo** and **has questions** lean legit (protective signals), meaning they are good “trust” indicators.

Important caveat:

Some features like **missing_title** / **missing_description** show inflated odds ratios despite being essentially never missing. These are artifacts of smoothing and should not be treated as meaningful signals.

E) Recurring scam “families” exist (campaign-like templates)

After segmenting fraud into a stricter scam-like subset and removing duplicates, clustering became much clearer and produced interpretable families.

My scam-like clustering output shows repeated templates such as:

1. Cash jobs / start immediately / free time template
2. WFH data entry / remote admin clerk template
3. “Totally free to join” + unrealistic daily earnings template
4. Smaller daily pay + phone-based conversion templates

3) Recommendation for a pilot detection system

Operating model: ALLOW / HOLD / BLOCK (tiered gating)

- ALLOW: low risk
- HOLD (Review/Verification): medium risk - highest value zone
- BLOCK: only for high-confidence cases

Detection design: Hybrid approach (rules + ML risk scoring)

Stage 1: High-confidence, explainable policy rules

Use the top feature bundles as reasons to HOLD/BLOCK (clear, auditable).

Examples based on your findings:

- Missing company profile + short text + number-heavy
- Missing company profile + pay-frequency framing
- Missing company profile + missing requirements/benefits
- Remote + pay-frequency + credibility gaps

Role: catch “obvious” risky patterns early with low false positives.

Stage 2 : Risk scoring model (generalize beyond exact rules)

- Logistic Regression over:
 - text representation (char/word n-grams)
 - plus my feature bank (missingness/completeness, digit ratio, contact/pay features, logo/questions)

Role: produce a calibrated risk score to rank cases and catch wording variations.

4) Risks & mitigations

Risks: false positives, mixed fraud types, scammer wording changes.

Mitigations: ALLOW/HOLD/BLOCK with strict BLOCK, keep scam-like subtyping, rely on robust signals (missingness/structure/conversion), monitor drift + new templates.

5) Pilot success criteria

Measure: BLOCK precision/False positive rate, recall at fixed precision, False positive per 10k postings

Validate: time-based validation (train on older ads, test on newer ads), Ensure duplicated ads don't appear in both train and test

